

Roofline Analysis

- Determine what is limiting kernel's performance

- Limited by 2 factors - whichever is hit first

- Compute (FLOPs/s) - how fast chip does math

- Memory (bytes/s) - how fast chip moves data

- Compute throughput: $\frac{W}{t}$ FLOPs/second

Kernel does W FLOPs

Memory throughput: $\frac{B}{t}$ bytes/second

Kernel moves B bytes from RAM

in t seconds for

some operation

- Limit: $\frac{W}{t} < \text{Peak FLOPs}$ - set by hardware

$\frac{B}{t} < \text{Peak Bandwidth}$

- Bottleneck: $t > \frac{W}{\text{Peak FLOPs}}$ $t > \frac{B}{\text{Peak Bandwidth}}$

time if compute
is bottleneck

time if memory
is bottleneck

Actual t will be max of $\frac{W}{\text{Peak FLOPs}}$ or $\frac{B}{\text{Peak Bandwidth}}$
as it has to be within constraints

Compare $\frac{W}{\text{Peak FLOPs}}$? $\frac{B}{\text{Peak Bandwidth}}$

$\frac{W}{B}$? $\frac{\text{Peak FLOPs}}{\text{Peak Bandwidth}}$

Arithmetic Intensity (AI)
of Kernel Algo

(BAL)

- Machine Balance
Peak arithmetic
intensity capable
on your hardware

- If $AI > BAL$ (more math per byte than hardware can do)

↳ Compute Bound

$AI < BAL$ (less math per byte than hardware can do)

↳ Memory Bound

$AI = BAL$

↳ Balanced

Performance

- What is kernel's achieved FLOPs/second $\frac{W}{T}$

- Compute bound where $T = \frac{W}{\text{Peak FLOPs/s}}$

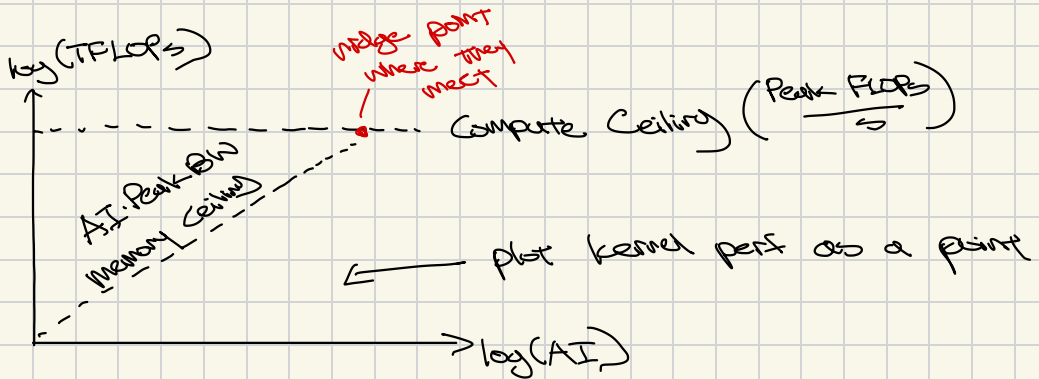
$$\text{Perf} = \frac{W}{T} = \frac{W}{\frac{W}{\text{Peak FLOPs/s}}} = \text{Peak FLOPs/s}$$

- Memory bound where $T = \frac{B}{\text{Peak Bandwidth}}$

$$\text{Perf} = \frac{W}{T} = \frac{W}{\frac{B}{\text{Peak Bandwidth}}} = \frac{W}{B} \cdot \text{Peak BW} = \text{AI} \cdot \text{Peak BW}$$

Roofline

Plo



Move right until ridge
Move up until compute

Left of ridge $\rightarrow$ memory bound
Right of ridge $\rightarrow$ compute bound

Note: memory can be referring to DRAM, L1, L2

Roofline can be based on any